

PACKAGE INSERT

1. NAME OF THE MEDICINAL PRODUCT

Candesartan HCT Sandoz 16mg/12.5mg Tablet

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each tablet contains 16 mg of candesartan cilexetil and 12.5 mg of hydrochlorothiazide.

Excipients with known effect: lactose monohydrate.

For the full list of excipients, see section 6.1 Excipients.

3. PHARMACEUTICAL FORM

Apricot, mottled, oval biconvex tablets, scored on both sides.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Hypertension when a monotherapy is not sufficiently effective.

4.2 Posology and method of administration

Dosage in hypertension

The recommended dose of Candesartan HCT Sandoz is one tablet once daily.

Most of the antihypertensive effect is usually attained within 4 weeks of initiation of treatment.

When clinically appropriate a direct change from monotherapy to Candesartan HCT Sandoz may be considered. Dose titration of candesartan cilexetil is recommended when switching from hydrochlorothiazide monotherapy.

Administration

For oral use.

Candesartan HCT Sandoz should be taken once daily with or without food.

Use in the elderly

No dosage adjustment is necessary in elderly patients.

Use in impaired renal function

In patients with mild to moderate renal impairment (i.e., creatinine clearance between 30-80ml/min/1.73 m² BSA), a dose titration is recommended.

Candesartan HCT Sandoz should not be used in patients with severe renal impairment (creatinine clearance <30 ml/min/1.73 m² BSA).

Use in impaired hepatic function

Patients with hepatic impairment: Dose titration is recommended in patients with mild to moderate chronic liver disease.

Candesartan HCT Sandoz should not be used in patients with severe hepatic impairment and/or cholestasis.

Use in children

The safety and efficacy of Candesartan HCT Sandoz have not been established in children.

4.3 Contraindications

- Hypersensitivity to any component of Candesartan HCT Sandoz or to sulfonamide derived drugs. (Hydrochlorothiazide is a sulfonamide derived drug)
- Pregnancy and lactation (see section 4.6)
- Severe renal impairment (creatinine clearance <30 ml/min/1.73 m² BSA)
- Severe hepatic impairment and/or cholestasis
- Gout
- Concomitant use of angiotensin II receptor antagonists (AIIRAs), including candesartan cilexetil, or of angiotensin converting enzyme (ACE) inhibitors with aliskiren in patients with diabetes mellitus (type I or II) or with moderate or severe renal impairment (GFR < 60ml/min/1.73m²) (see section 4.5).

4.4 Special warnings and precautions for use

Dual blockade of the renin-angiotensin-aldosterone system (RAAS)

There is evidence that the concomitant use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren

increases the risk of hypotension, hyperkalaemia and decreased renal function (including acute renal failure). Dual blockade of RAAS through the combined use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren is therefore not recommended (see section 4.5).

If dual blockade therapy is considered absolutely necessary, this should only occur under specialist supervision and subject to frequent close monitoring of renal function, electrolytes and blood pressure.

ACE-inhibitors and angiotensin II receptor blockers should not be used concomitantly in patients with diabetic nephropathy.

The use of candesartan cilexetil with aliskiren is contraindicated in patients with diabetes mellitus (type I or II) or moderate to severe renal impairment (GFR <60 ml/min/1.73m²) (see section 4.3).

Renal impairment

As with other agents inhibiting the renin-angiotensin-aldosterone system, changes in renal function may be anticipated in susceptible patients treated with candesartan cilexetil/hydrochlorothiazide (see section 4.3).

Kidney transplantation

There is limited clinical evidence regarding candesartan cilexetil/hydrochlorothiazide use in patients who have undergone renal transplant.

Renal artery stenosis

Other drugs that affect the renin-angiotensin-aldosterone system (RAAS), including angiotensin converting enzyme (ACE) inhibitors, may increase blood urea and serum creatinine in patients with bilateral renal artery stenosis or stenosis of the artery to a solitary kidney. A similar effect may be anticipated with angiotensin II receptor antagonists.

Intravascular volume depletion

In patients with intravascular volume and/or sodium depletion symptomatic hypotension may occur, as described for other agents acting on the RAAS. Therefore, the use of Candesartan HCT Sandoz is not recommended until this condition has been corrected.

Anaesthesia and surgery

Hypotension may occur during anaesthesia and surgery in patients treated with AIIRAs due to blockade of the renin-angiotensin system. Very rarely, hypotension may be severe such that it may warrant the use of intravenous fluids and/or vasopressors.

Hepatic impairment

Thiazides should be used with caution in patients with impaired hepatic function or progressive liver disease, since minor alterations of fluid and electrolyte balance may precipitate hepatic coma. There is no clinical experience with Candesartan HCT Sandoz in patients with hepatic impairment.

Aortic and mitral valve stenosis or obstructive hypertrophic cardiomyopathy

As with other vasodilators, special caution is indicated in patients suffering from haemodynamically relevant aortic or mitral valve stenosis, or obstructive hypertrophic cardiomyopathy.

Primary hyperaldosteronism

Patients with primary hyperaldosteronism generally will not respond to antihypertensive agents acting through inhibition of the RAAS. Therefore the use of Candesartan HCT Sandoz is not recommended in this population.

Electrolyte imbalance

As for any patient receiving diuretic therapy, periodic determination of serum electrolytes should be performed at appropriate intervals.

Thiazides, including hydrochlorothiazide, can cause fluid or electrolyte imbalance (hypercalcaemia, hypokalaemia, hyponatraemia, hypomagnesaemia and hypochloraemic alkalosis).

Hydrochlorothiazide dose-dependently increases urinary potassium excretion which may result in hypokalaemia. This effect of hydrochlorothiazide seems to be less evident when combined with candesartan cilexetil.

Based on experience with the use of other drugs that affect the renin-angiotensin-aldosterone system, concomitant use of Candesartan HCT Sandoz and ACE inhibitors, aliskiren, potassium-sparing diuretics, potassium supplements or salt substitutes or other drugs that may increase serum potassium levels (e.g. heparin, co- trimoxazole) may lead to increases in serum potassium.

Metabolic and endocrine effects

Treatment with a thiazide diuretic may impair glucose tolerance. Dose adjustment of antidiabetic medicinal products, including insulin, may be required. Latent diabetes mellitus may become manifest during thiazide therapy. Increases in cholesterol and triglyceride levels have been associated with thiazide diuretic therapy. However, at the 12.5mg contained Candesartan HCT Sandoz, minimal or no effects were reported. Thiazide diuretics increase serum uric acid concentration and may precipitate gout in susceptible patients.

Photosensitivity

Cases of photosensitivity reactions have been reported during use of thiazide diuretics (see section 4.8). If a photosensitivity reaction occurs, it is recommended to stop treatment. If re-administration of treatment is essential, it is recommended to protect areas exposed to the sun or to artificial UVA radiation.

Choroidal effusion, Acute myopia and secondary angle-closure glaucoma

Hydrochlorothiazide, a sulfonamide, can cause an idiosyncratic reaction resulting in choroidal effusion with visual field defect, transient myopia and acute angle-closure glaucoma. Symptoms include acute onset of decreased visual acuity or ocular pain and typically occur within hours to weeks of medicinal product initiation. Untreated acute angle-closure glaucoma can lead to permanent vision loss.

The primary treatment is to discontinue drug intake as rapidly as possible. Prompt medical or surgical treatment may need to be considered if the intraocular pressure remains uncontrolled. Risk factors for developing acute angle-closure glaucoma may include a history of sulfonamides or penicillin allergy.

General

In patients whose vascular tone and renal function depend predominantly on the activity of the RAAS (e.g. patients with severe congestive heart failure or underlying renal disease, including renal artery stenosis), treatment with other drugs that affect this system has been associated with acute hypotension, azotaemia, oliguria or, rarely, acute renal failure. As with any antihypertensive agent, excessive blood pressure decrease in patients with ischaemic heart disease or atherosclerotic cerebrovascular disease could result in a myocardial infarction or stroke.

Hypersensitivity reactions to hydrochlorothiazide may occur in patients with or without a history of allergy or bronchial asthma, but are more likely in patients with such a history.

Exacerbation or activation of systemic lupus erythematosus has been reported with the use of thiazide diuretics.

The antihypertensive effect of Candesartan HCT Sandoz may be enhanced by other antihypertensives.

This medicinal product contains lactose, as an excipient, and patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Non-melanoma skin cancer

An increased risk of non-melanoma skin cancer (NMSC) [basal cell carcinoma (BCC) and squamous cell carcinoma (SCC)] with increasing cumulative dose of hydrochlorothiazide (HCTZ) exposure has been observed in two epidemiological studies based on the Danish National Cancer Registry. Photosensitizing actions of HCTZ could act as a possible mechanism for NMSC.

Patients taking HCTZ should be informed of the risk of NMSC and advised to regularly check their skin for any new lesions and promptly report any suspicious skin lesions. Possible preventive measures such as limited exposure to sunlight and UV rays and, in case of exposure, adequate protection should be advised to the patients in order to minimize the risk of skin cancer. Suspicious skin lesions should be promptly examined potentially including histological examinations of biopsies. The use of HCTZ may also need to be reconsidered in patients who have experienced previous NMSC.

Acute respiratory toxicity

Very rare severe cases of acute respiratory toxicity, including acute respiratory distress syndrome (ARDS) have been reported after taking hydrochlorothiazide. Pulmonary oedema typically develops within minutes to hours after hydrochlorothiazide intake. At the onset, symptoms include dyspnoea, fever, pulmonary deterioration, and hypotension. If diagnosis of ARDS is suspected, candesartan cilexetil/hydrochlorothiazide should be withdrawn, and appropriate treatment given. Hydrochlorothiazide should not be administered to patients who previously experienced ARDS following hydrochlorothiazide intake.

4.5 Interaction with other medicinal products or other forms of interaction

Compounds which have been investigated with candesartan cilexetil in clinical pharmacokinetic studies include hydrochlorothiazide, warfarin, digoxin, oral contraceptives (i.e. ethinylestradiol/levonorgestrel), glibenclamide and nifedipine. No pharmacokinetic interactions of clinical significance were identified in these studies.

The bioavailability of candesartan is not affected by food.

The potassium depleting effect of hydrochlorothiazide could be expected to be potentiated by other medicinal products associated with potassium loss and hypokalaemia (e.g. other kaliuretic diuretics, laxatives, amphotericin, carbenoxolone, salicylic acid derivatives).

Concomitant use of Candesartan HCT Sandoz and potassium-sparing diuretics, potassium supplements or salt substitutes or other medicinal products that may increase serum potassium levels (e.g. heparin sodium, co-trimoxazole also known as trimethoprim/sulfamethoxazole) may lead to increases in serum potassium. Monitoring of potassium should be undertaken as appropriate (see section 4.4).

Diuretic-induced hypokalaemia and hypomagnesaemia predisposes to the potential cardiotoxic effects of digitalis glycosides and antiarrhythmics. Periodic monitoring of serum potassium is recommended when Candesartan HCT Sandoz is administered with such medicinal products, and with the following medicinal products that could induce *torsades de pointes*:

- Class Ia antiarrhythmics (e.g. quinidine, hydroquinidine, disopyramide)
- Class III antiarrhythmics (e.g. amiodarone, sotalol, dofetilide, ibutilide)
- Some antipsychotics (e.g. thioridazine, chlorpromazine, levomepromazine, trifluoperazine, cyamemazine, sulpiride, sultopride, amisulpride, tiapride, pimozide, haloperidol, droperidol)
- Others (e.g. bepridil, cisapride, diphemanil, erythromycin iv, halofantrin, ketanserin, mizolastin, pentamidine, sparfloxacin, terfenadine, vincamine iv)

Reversible increases in serum lithium concentrations and toxicity have been reported during concomitant administration of lithium with ACE inhibitors or hydrochlorothiazide. A similar effect has also been reported with AIIIRAs. Use of Candesartan HCT Sandoz with lithium is not recommended. If the combination proves necessary, careful monitoring of serum lithium levels is recommended.

When AIIIRAs are administered simultaneously with non-steroidal anti-inflammatory drugs (NSAIDs) (i.e. selective COX-2 inhibitors, acetylsalicylic acid (>3 g/day) and non-selective NSAIDs), attenuation of the antihypertensive effect may occur.

As with ACE inhibitors, concomitant use of AIIIRAs and NSAIDs may lead to an increased risk of worsening of renal function, including possible acute renal failure, and an increase in serum potassium, especially in patients with poor pre-existing renal function. The combination should be administered with caution, especially in the older patients and in volume-depleted patients. Patients should be adequately hydrated and consideration should be given to monitoring renal function after initiation of concomitant therapy, and periodically thereafter.

The diuretic, natriuretic and antihypertensive effect of hydrochlorothiazide is blunted by NSAIDs.

The absorption of hydrochlorothiazide is reduced by colestipol or cholestyramine.

The effect of nondepolarising skeletal muscle relaxants (e.g. tubocurarine) may be potentiated by hydrochlorothiazide.

Thiazide diuretics may increase serum calcium levels due to decreased excretion. If calcium supplements or Vitamin D must be prescribed, serum calcium levels should be monitored and the dose adjusted accordingly.

The hyperglycaemic effect of beta-blockers and diazoxide may be enhanced by thiazides.

Anticholinergic agents (e.g. atropine, biperiden) may increase the bioavailability of thiazide-type diuretics by decreasing gastrointestinal motility and stomach emptying rate.

Thiazide may increase the risk of adverse effects caused by amantadine.

Thiazides may reduce the renal excretion of cytotoxic medicinal products (e.g. cyclophosphamide, methotrexate) and potentiate their myelosuppressive effects.

Postural hypotension may become aggravated by simultaneous intake of alcohol, barbiturates or anaesthetics.

Treatment with a thiazide diuretic may impair glucose tolerance. Dose adjustment of antidiabetic medicinal products, including insulin, may be required. Metformin should be used with caution because of the risk of lactic acidosis induced by possible functional renal failure linked to hydrochlorothiazide.

Hydrochlorothiazide may decrease arterial responsiveness to noradrenaline but not enough to preclude effectiveness of the pressor agent for therapeutic use.

Hypokalemia may develop during concomitant use of steroids or adrenocorticotrophic hormone (ACTH).

There is no clinically significant interaction between hydrochlorothiazide and food.

Hydrochlorothiazide may increase the risk of acute renal insufficiency especially with high doses of iodinated contrast media.

Concomitant treatment with ciclosporin may increase the risk of hyperuricaemia and gout-type complications.

Concomitant treatment with baclofen, amifostin, tricyclic antidepressants or neuroleptics may lead to enhancement of the antihypertensive effect and may induce hypotension.

Dual blockade of the RAAS with AIIRAs, ACE inhibitors, or aliskiren

The concomitant use of AIIRAs, including candesartan cilexetil, with other agents acting on the RAAS is associated with an increased incidence of hypotension, hyperkalaemia, and changes in renal function compared to monotherapy. It is recommended to monitor blood pressure, renal function and electrolytes in patients on candesartan cilexetil and other agents that affect the RAAS (see section 4.4).

The concomitant use of AIIRAs, including candesartan cilexetil, or of ACE inhibitors with aliskiren, should be avoided in patients with severe renal impairment (GFR <30ml/min) (see section 4.4).

The concomitant use of AIIRAs, including candesartan cilexetil, or ACE inhibitors with aliskiren is contraindicated in patients with type 2 diabetes (see section 4.3).

4.6 Pregnancy and lactation

Use in pregnancy

The use of Candesartan HCT Sandoz is contraindicated during pregnancy (see section 4.3). Patients receiving Candesartan HCT Sandoz should be made aware of that before contemplating a possibility of becoming pregnant so that they can discuss appropriate options with their treating physician. When pregnancy is diagnosed, treatment with Candesartan HCT Sandoz must be stopped immediately and if appropriate, alternative therapy should be started.

When used in pregnancy, drugs that act directly on the renin-angiotensin system can cause foetal and neonatal injury and death. Exposure to angiotensin II receptor antagonist therapy is known to induce human fetotoxicity (decreased renal function, oligohydramnios, skull ossification retardation) and neonatal toxicity (renal failure, hypotension, hyperkalaemia).

There is limited experience with hydrochlorothiazide during pregnancy, especially during the first trimester. Animal studies are insufficient. Hydrochlorothiazide crosses the placenta. Based on the pharmacological mechanism of action of hydrochlorothiazide, its use during pregnancy may compromise foeto-placental perfusion and may cause foetal and neonatal effects like icterus, disturbance of electrolyte balance and thrombocytopenia.

Use in lactation

It is not known whether candesartan is excreted in human milk. However, candesartan is excreted in the milk of lactating rats. Hydrochlorothiazide passes into mother's milk.

Because of the potential for adverse effects on the nursing infant, breast feeding should be discontinued if the use of Candesartan HCT Sandoz is considered essential (see section 4.3).

4.7 Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed, but based on its pharmacodynamic properties Candesartan HCT Sandoz is unlikely to affect this ability. When driving vehicles or operating machines, it should be taken into account that occasionally dizziness or weariness may occur during treatment with Candesartan HCT Sandoz.

4.8 Undesirable effects

Candesartan cilexetil

System Organ Class	Frequency	Undesirable Effect
Infections and infestations	Common	Respiratory infection
Blood and lymphatic system disorders	Very rare	Leukopenia, neutropenia and agranulocytosis
Metabolism and nutrition disorders	Very rare	Hyperkalaemia, hyponatraemia
Nervous system disorders	Very rare	Dizziness/vertigo, headache

Respiratory, thoracic and mediastinal disorders	Very rare	Cough
Gastrointestinal disorders	Very rare	Nausea
	Not known	Diarrhea
Hepatobiliary disorders	Very rare	Increased liver enzymes, abnormal hepatic function or hepatitis
Skin and subcutaneous tissue disorders	Very rare	Angioedema, rash, urticaria, pruritus
Musculoskeletal and connective tissue disorders	Very rare	Back pain, arthralgia, myalgia
Renal and urinary disorders	Very rare	Renal impairment, including renal failure in susceptible patients (see section 4.4)

Hydrochlorothiazide

The table below presents adverse reactions with hydrochlorothiazide monotherapy usually with doses of 25 mg or higher.

System Organ Class	Frequency	Undesirable Effect
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	Not known	Non-melanoma skin cancer (basal cell carcinoma and squamous cell carcinoma) <u>Description of selected adverse reactions</u> Non-melanoma skin cancer: Based on available data from epidemiological studies, cumulative dose-dependent association between HCTZ and NMSC has been observed.
Blood and lymphatic system disorders	Rare	Leukopenia, neutropenia/agranulocytosis, thrombocytopenia, aplastic anaemia, bone marrow depression, haemolytic anaemia
Immune system disorders	Rare	Anaphylactic reactions
Metabolism and nutrition disorders	Common	Hyperglycaemia, hyperuricaemia, electrolyte imbalance (including hyponatraemia and hypokalaemia)
Psychiatric disorders	Rare	Sleep disturbances, depression, restlessness
Nervous system disorders	Common	Light-headedness, vertigo
	Rare	Paraesthesia
Eye disorders	Rare	Transient blurred vision
	Not known	Choroidal effusion, acute myopia, acute angle-closure glaucoma
Cardiac disorders	Rare	Cardiac arrhythmias
Vascular disorders	Uncommon	Postural hypotension
	Rare	Necrotising angitis (vasculitis, cutaneous vasculitis)
Respiratory, thoracic and mediastinal disorders	Rare	Respiratory distress (including pneumonitis and pulmonary oedema)
	Very rare	Acute respiratory distress syndrome (ARDS)
Gastrointestinal disorders	Uncommon	Anorexia, loss of appetite, gastric irritation, diarrhoea, constipation
	Rare	Pancreatitis
Hepatobiliary disorders	Rare	Jaundice (intrahepatic cholestatic jaundice)
Skin and subcutaneous tissue disorders	Uncommon	Rash, urticaria, photosensitivity reactions
	Rare	Toxic epidermal necrolysis
	Not known	Systemic lupus erythematosus, cutaneous lupus erythematosus
Musculoskeletal and connective tissue disorders	Rare	Muscle spasm
Renal and urinary disorders	Common	Glycosuria
	Rare	Renal dysfunction and interstitial nephritis
General disorders and administration site conditions	Common	Weakness
	Rare	Fever
Investigations	Common	Increases in cholesterol and triglycerides
	Rare	Increases in BUN and serum creatinine

Laboratory findings

In general, there were no clinically important influences of candesartan cilexetil/hydrochlorothiazide on routine laboratory variables. Increases in serum uric acid, blood glucose and serum ALAT (SGPT) were reported as adverse events slightly more often with candesartan cilexetil/hydrochlorothiazide (crude rates 1.1%, 1.0% and 0.9%, respectively) than with placebo (0.4%, 0.2% and 0%, respectively). Minor decreases in haemoglobin and

increases in serum ASAT (SGOT) have been observed in single patients receiving candesartan cilexetil/hydrochlorothiazide. Increases in creatinine, urea or potassium and decrease in sodium have been observed.

4.9 Overdose

Symptoms

Based on pharmacological considerations, the main manifestation of an overdose of candesartan cilexetil is likely to be symptomatic hypotension and dizziness. In individual case reports of overdose (of up to 672 mg candesartan cilexetil) patient recovery was uneventful.

The main manifestation of an overdose of hydrochlorothiazide is acute loss of fluid and electrolytes. Symptoms such as dizziness, hypotension, thirst, tachycardia, ventricular arrhythmias, sedation/impairment of consciousness and muscle cramps can also be observed.

Management

No specific information is available on the treatment of overdose with Candesartan HCT Sandoz. The following measures are, however, suggested in case of overdose.

When indicated, induction of vomiting or gastric lavage should be considered. If symptomatic hypotension should occur, symptomatic treatment should be instituted and vital signs monitored. The patient should be placed supine with the legs elevated. If this is not sufficient, plasma volume should be increased by infusion of isotonic saline solution. Serum electrolyte and acid balance should be checked and corrected, if needed. Sympathomimetic medicinal products may be administered if the above-mentioned measures are not sufficient.

Candesartan cannot be removed by haemodialysis. It is not known to what extent hydrochlorothiazide is removed by haemodialysis.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmaco-therapeutic group: Angiotensin II antagonists + diuretics, ATC code: C09DA06

Angiotensin II is the primary vasoactive hormone of the RAAS and plays a role in the pathophysiology of hypertension and other cardiovascular disorders.

It also has a role in the pathogenesis of organ hypertrophy and end organ damage. The major physiological effects of angiotensin II, such as vasoconstriction, aldosterone stimulation, regulation of salt and water homeostasis and stimulation of cell growth, are mediated via the type 1(AT₁) receptor.

Candesartan cilexetil is a prodrug which is rapidly converted to the active drug, candesartan, by ester hydrolysis during absorption from the gastrointestinal tract. Candesartan is an AIIRA, selective for AT₁ receptors, with tight binding to and slow dissociation from the receptor. It has no agonist activity.

Candesartan does not influence ACE or other enzyme systems usually associated with the use of ACE inhibitors. Since there is no effect on the degradation of kinins, or on the metabolism of other substances, such as substance P, AIIRAs are unlikely to be associated with cough. In controlled clinical trials comparing candesartan cilexetil with ACE inhibitors, the incidence of cough was lower in patients receiving candesartan cilexetil. Candesartan does not bind to or block other hormone receptors or ion channels known to be important in cardiovascular regulation. The antagonism of the AT₁ receptors results in dose related increases in plasma renin levels, angiotensin I and angiotensin II levels, and a decrease in plasma aldosterone concentration.

Hydrochlorothiazide inhibits the active reabsorption of sodium, mainly in the distal kidney tubules, and promotes the excretion of sodium, chloride and water. The renal excretion of potassium and magnesium increases dose-dependently, while calcium is reabsorbed to a greater extent. Hydrochlorothiazide decreases plasma volume and extracellular fluid and reduces cardiac output and blood pressure. During long-term therapy, reduced peripheral resistance contributes to the blood pressure reduction.

Large clinical studies have shown that long-term treatment with hydrochlorothiazide reduces the risk for cardiovascular morbidity and mortality.

Candesartan and hydrochlorothiazide have additive antihypertensive effects.

In hypertensive patients, candesartan cilexetil/hydrochlorothiazide results in a dose-dependent and long-lasting reduction in arterial blood pressure without reflex increase in heart rate. There is no indication of serious or exaggerated first dose hypotension or rebound effect after cessation of treatment. After administration of a single dose of candesartan cilexetil/hydrochlorothiazide, onset of the antihypertensive effect generally occurs within 2 hours. With continuous treatment, most of the reduction in blood pressure is attained within four weeks and is

sustained during long-term treatment. Candesartan cilexetil/hydrochlorothiazide once daily provides effective and smooth blood pressure reduction over 24 hours, with little difference between maximum and trough effects during the dosing interval.

In double-blind, randomised studies, the incidence of adverse events, especially cough, was lower during treatment with candesartan cilexetil/hydrochlorothiazide than during treatment with combinations of ACE inhibitors and hydrochlorothiazide.

Candesartan cilexetil/hydrochlorothiazide is similarly effective in patients irrespective of age and gender.

Currently there are no data on the use of candesartan cilexetil/hydrochlorothiazide in patients with renal disease/nephropathy, reduced left ventricular function/congestive heart failure and post myocardial infarction.

Non-melanoma skin cancer

Based on available data from epidemiological studies, cumulative dose-dependent association between HCTZ and NMSC has been observed. One study included a population comprised of 71,533 cases of BCC and of 8,629 cases of SCC matched to 1,430,833 and 172,462 population controls, respectively. High HCTZ use ($\geq 50,000$ mg cumulative) was associated with an adjusted OR of 1.29 (95% CI: 1.23-1.35) for BCC and 3.98 (95% CI: 3.68-4.31) for SCC. A clear cumulative dose response relationship was observed for both BCC and SCC. Another study showed a possible association between lip cancer (SCC) and exposure to HCTZ: 633 cases of lip-cancer were matched with 63,067 population controls, using a risk-set sampling strategy. A cumulative dose-response relationship was demonstrated with an adjusted OR 2.1 (95% CI: 1.7-2.6) increasing to OR 3.9 (3.0-4.9) for high use ($\sim 25,000$ mg) and OR 7.7 (5.7-10.5) for the highest cumulative dose ($\sim 100,000$ mg).

5.2 Pharmacokinetic properties

Concomitant administration of candesartan cilexetil and hydrochlorothiazide has no clinically significant effect on the pharmacokinetics of either medicinal product.

Absorption and distribution

Candesartan cilexetil

Following oral administration, candesartan cilexetil is converted to the active substance candesartan. The absolute bioavailability of candesartan is approximately 40% after an oral solution of candesartan cilexetil.

The relative bioavailability of a tablet formulation of candesartan cilexetil compared with the same oral solution is approximately 34% with very little variability. The mean peak serum concentration (C_{max}) is reached 3-4 hours following tablet intake. The candesartan serum concentrations increase linearly with increasing doses in the therapeutic dose range. No gender related differences in the pharmacokinetics of candesartan have been observed. The area under the serum concentration versus time curve (AUC) of candesartan is not significantly affected by food.

Candesartan is highly bound to plasma protein (more than 99%). The apparent volume of distribution of candesartan is 0.1 l/kg.

Hydrochlorothiazide

Hydrochlorothiazide is rapidly absorbed from the gastrointestinal tract with an absolute bioavailability of approximately 70%. Concomitant intake of food increases the absorption by approximately 15%. The bioavailability may decrease in patients with cardiac failure and pronounced oedema.

The plasma protein binding of hydrochlorothiazide is approximately 60%. The apparent volume of distribution is approximately 0.8 l/kg.

Metabolism and elimination

Candesartan cilexetil

Candesartan is mainly eliminated unchanged via urine and bile and only to a minor extent eliminated by hepatic metabolism (CYP2C9). Available interaction studies indicate no effect on CYP2C9 and CYP3A4.

Based on *in vitro* data, no interaction would be expected to occur *in vivo* with medicinal products whose metabolism is dependent upon cytochrome P450 isoenzymes CYP1A2, CYP2A6, CYP2C9, CYP2C19, CYP2D6, CYP2E1 or CYP3A4. The terminal half-life ($t_{1/2}$) of candesartan is approximately 9 hours. There is no accumulation following multiple doses. The half-life of candesartan remains unchanged (approximately 9 h) after administration of candesartan cilexetil in combination with hydrochlorothiazide. No additional accumulation of candesartan occurs after repeated doses of the combination compared to monotherapy.

Total plasma clearance of candesartan is about 0.37 ml/min/kg, with a renal clearance of about 0.19 ml/min/kg. The renal elimination of candesartan is both by glomerular filtration and active tubular secretion. Following an oral dose of ^{14}C -labelled candesartan cilexetil, approximately 26% of the dose is excreted in the urine as candesartan

and 7% as an inactive metabolite while approximately 56% of the dose is recovered in the faeces as candesartan and 10% as the inactive metabolite.

Hydrochlorothiazide

Hydrochlorothiazide is not metabolised and is excreted almost entirely as unchanged drug by glomerular filtration and active tubular secretion. The terminal $t_{1/2}$ of hydrochlorothiazide is approximately 8 hours. Approximately 70% of an oral dose is eliminated in the urine within 48 hours. The half-life of hydrochlorothiazide remains unchanged (approximately 8 h) after administration of hydrochlorothiazide in combination with candesartan cilexetil. No additional accumulation of hydrochlorothiazide occurs after repeated doses of the combination compared to monotherapy.

Pharmacokinetics in special populations

Candesartan cilexetil

In elderly subjects (over 65 years), C_{max} and AUC of candesartan are increased by approximately 50% and 80%, respectively in comparison to young subjects. However, the blood pressure response and the incidence of adverse events are similar after a given dose of candesartan cilexetil/hydrochlorothiazide in young and elderly patients (see section 4.2).

In patients with mild to moderate renal impairment, C_{max} and AUC of candesartan increased during repeated dosing by approximately 50% and 70%, respectively, but the terminal $t_{1/2}$ was not altered, compared to patients with normal renal function. The corresponding changes in patients with severe renal impairment were approximately 50% and 110%, respectively. The terminal $t_{1/2}$ of candesartan was approximately doubled in patients with severe renal impairment. The pharmacokinetics in patients undergoing haemodialysis were similar to those in patients with severe renal impairment.

In patients with mild to moderate hepatic impairment, there was an increase in the AUC of candesartan of approximately 20%. In patients with moderate to severe hepatic impairment, the increase in the AUC of candesartan was approximately 80%.

Hydrochlorothiazide

The terminal $t_{1/2}$ of hydrochlorothiazide is prolonged in patients with renal impairment.

6. PHARMACEUTICAL PARTICULARS

6.1 List of Excipients

Lactose monohydrate
Maize starch
Povidone K30
Carrageenan
Croscarmellose sodium
Magnesium stearate
Purified water
Ferric oxide red
Ferric oxide yellow

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

2 years.

6.4 Special precautions for storage

Do not store above 30°C.

Store in the original package in order to protect from moisture.

6.5 Dosage form and packaging available

Candesartan HCT Sandoz 16mg/12.5mg is packed in the Al/Al blister with desiccant. Packing components include coldform foil (aluminium, soft, unprinted foil with desiccant) and lidding foil (hard aluminium foil).

There are two registered pack sizes:

28 tablets per box - 7 tablets are packed in the Al/Al blister and 4 blisters (4x7's) are packed into an outer carton folding box.

30 tablets per box - 10 tablets are packed in the Al/Al blister and 3 blisters (3x10's) are packed into an outer carton folding box.

Not all pack sizes are marketed.

7. PRODUCT REGISTRATION HOLDER

Sandoz Products Malaysia Sdn. Bhd.
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